ABHISHEK KUMAR

FULL STACK ENGINEER

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Portfolio: abhishek-portfolio.vercel.app | GitHub: abhishekshakya1 | LinkedIn: abhishekshakya1 | LeetCode: 116+ Solved

PROFESSIONAL SUMMARY

Full Stack Engineer and BCA ('26) scholar with hands-on capability architecting high-throughput distributed systems, event ingestion pipelines, and production MERN architectures. Creator of **PulseEngine** (sub-3ms Redis telemetry stream, Gemini 3.8 Flash AI SRE) and **CareConnect** (3-tier clinic booking with 401 Axios token rotation). Solved **116+ algorithmic challenges on LeetCode** in JavaScript with deep focus on time-space complexity optimization. Proficient in layered design (**Controller → Service → Repository**), JWT session rotation, atomic database mutations, and containerized deployments across **Vercel, Render, and Docker**.

TECHNICAL SKILLS

Programming Languages:	JavaScript (ES6+), TypeScript, C++, Java, HTML5, CSS3, SQL
Frontend Architecture:	React.js, Next.js 16 (App Router), Tailwind CSS, Framer Motion, Axios, State Management
Backend & Distributed:	Node.js, Express.js, RESTful API Design, JWT Session Rotation, RBAC, Redis Queuing, Rate Limiting
Databases & ORM:	PostgreSQL (Neon Serverless), MongoDB (Mongoose), Redis (Upstash), Prisma ORM, Normalization
DevOps, AI & Cloud Tooling:	Google Gemini AI API, Docker, Git, GitHub, Postman, Vercel, Render, Polar.sh SDK, Zod, Linux
Core CS Fundamentals:	Data Structures & Algorithms, Object-Oriented Programming (OOP), System Architecture, DBMS

TECHNICAL PROJECTS & EXPERIENCE

PulseEngine – High-Throughput Telemetry Pipeline & Autonomous AI SRE — 2026

Next.js 16, TypeScript, Upstash Redis, Neon PostgreSQL, Prisma, Google Gemini AI, Docker, Polar.sh | Live Demo | GitHub

- Architected a distributed telemetry ingestion pipeline processing event streams with **sub-3ms latency**, leveraging an in-memory Redis buffer (`LPUSH`) to prevent database connection pool exhaustion under high concurrency.
- Engineered an asynchronous worker daemon handling micro-batch persistence (**500 events/batch**) into Neon Serverless PostgreSQL via Prisma ORM, scaling throughput up to **100k+ events/sec**.
- Integrated **Google Gemini 3.8 Flash** to build an autonomous AI SRE Copilot executing real-time root-cause analysis (RCA), blast radius evaluation, and generating copyable 1-click remediation shell commands upon SLA breaches.
- Implemented **Polar.sh subscription billing** with automated webhook synchronization, tenant quota metering, and dynamic soft-cap rate limiting; containerized the engine with multi-stage Docker builds.

CareConnect – Healthcare Consultation & Clinic Management Platform — 2025 - 2026

React.js, Node.js, Express.js, MongoDB, Tailwind CSS, Zod, JWT, Cloudinary, Vercel, Render | Live Demo | GitHub

- Architected a 3-tier MERN platform (**Controller → Service → Repository**) serving 13 modular REST endpoints for doctor discovery, atomic appointment scheduling, and patient management.
- Engineered an **Axios response interceptor queue pipeline** that intercepts 401 statuses, coordinates token rotation, and replays enqueued requests, eliminating **100% of concurrent session drops**.
- Implemented atomic MongoDB appointment slot reservations preventing concurrent double-booking, enforced Role-Based Access Control (RBAC) with HTTP-only JWTs, and sanitized inputs via Zod schemas.
- Integrated multipart media uploads via Multer and Cloudinary SDK; streamlined diagnostics using Winston and Morgan, deploying backend APIs to Render.

Interactive Developer Portfolio & System Architecture Showcase — 2026

React 19, Tailwind CSS v4, Framer Motion, Web Audio API, Vite, Vercel | Live Demo | GitHub

- Engineered an interactive portfolio achieving a **98+ Google Lighthouse performance score**, featuring interactive 3D physics cards, dynamic theme customization, and zero-asset browser Web Audio API sound synthesis.
- Implemented instant zero-JS obsidian preloading and `<link rel="preload">` asset pipelines with skeleton placeholders, achieving **< 150ms** first contentful paint (FCP).

EDUCATION

Bachelor of Computer Applications (BCA) — 2023 - 2026

DAV Centenary College, Faridabad • Affiliated to **Maharshi Dayanand University (MDU), Rohtak** | **Aggregate: 71.03%**

Relevant Coursework: Data Structures & Algorithms, Database Management Systems (SQL), Operating Systems, Computer Networks, Software Engineering, Object-Oriented Programming (Java/C++).

Senior Secondary (Class XII) - Science Stream — HBSE Board | Score: 77.2%

Govt. Model Sanskriti Senior Secondary School (GMSSSS)

Secondary (Class X) - Matriculation — HBSE Board | Score: 92.8%

Govt. Model Sanskriti Senior Secondary School (GMSSSS)

CERTIFICATIONS & ALGORITHMIC CREDENTIALS

- **LeetCode Algorithmic Milestones:** Completed **116+ algorithmic challenges in JavaScript** (Profile Link) across Arrays, Two Pointers, Sliding Window, Graphs, and DP Optimization.
- **Full Stack Web Development 1.0:** Physics Wallah Skills (Credential Link) — Comprehensive curriculum in React.js, Node.js, Express, MongoDB, and REST API Architecture (Dec 2024).
- **Namaste DSA:** NamasteDev by Akshay Saini (Credential Link) — Rigorous training on Data Structures, Algorithms, Time-Space Complexity Analysis, and algorithmic design patterns.